



A simple electrochemical indicator of lithium plating based on current response at the constant current to constant voltage charging transition

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HIGHLIGHTS

- A non-invasive lithium plating indicator γ is proposed using CC–CV kinetics.
- γ detects plating about 50 cycles before rapid capacity fade begins.
- Physics-based modeling shows the indicator arises from current redistribution.
- The method requires no additional hardware or prolonged rest periods.
- Robustness is validated with open datasets and fast-charging protocols.

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ABSTRACT

Lithium plating critically degrades lithium-ion batteries and poses safety risks, especially during fast charging. However, practical non-invasive detection remains challenging. We propose a novel diagnostic approach based on transient current kinetics during the constant-current (CC) to constant-voltage (CV) transition. A dimensionless indicator, γ , is introduced to quantify current decay immediately following the CC–CV switch. By employing a physics-based electrochemical model coupled with side reactions, we elucidate the mechanistic origin of this indicator: it captures the abrupt kinetic redistribution and the inheritance of suppressed intercalation currents caused by the cessation of plating under voltage control. Simulations demonstrate that γ exhibits a distinct trend reversal precisely at the onset of lithium plating. This signature provides an early warning approximately 50 cycles before noticeable capacity divergence, offering superior sensitivity compared to voltage relaxation analysis. The robustness of the proposed method is systematically verified across various C-rates (0.2C–1.5C) and complex multi-step CC–CV protocols. Furthermore, the method is validated using independent open-source experimental datasets, confirming its general applicability. Requiring no additional hardware or prolonged rest periods, this strategy offers a cost-effective, real-time solution for enhancing battery safety.

1. Introduction

With the growing demand for high-energy-density lithium-ion batteries (LIBs) in electric vehicles and grid-scale energy storage systems, ensuring their long-term cycling safety has become a critical priority [1, 2]. Among various degradation and failure mechanisms, lithium plating represents one of the most severe risks. This phenomenon, in which metallic lithium deposits on the anode surface rather than intercalating

into it, not only accelerates capacity degradation but also significantly increases the likelihood of internal short circuits and thermal runaway [3,4]. Therefore, the development of reliable and early-stage detection methods for lithium plating is essential for effective battery safety management and lifetime prediction [5,6].

To date, numerous techniques have been proposed for detecting lithium plating. Ex situ and *in situ* imaging methods—such as scanning electron microscopy (SEM) [7], optical microscopy [8], and X-ray or

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neutron imaging [9]—provide direct visual evidence of metallic lithium but are inherently destructive or require complex cell configurations, thereby limiting their practical applicability [10,11]. Nuclear magnetic resonance (NMR) spectroscopy [12] can quantitatively distinguish metallic lithium from intercalated species; however, its high cost and operational complexity hinder large-scale implementation [13]. Electrochemical impedance spectroscopy (EIS) offers valuable insights into interfacial changes during plating, yet it is time-consuming and requires additional measurement and signal-processing modules under dynamic conditions [14–16]. Voltage response-based methods [17], including differential voltage (dV/dQ) [18] and incremental capacity (dQ/dV) analyses [19], have shown promise in identifying the onset of lithium plating but typically demand long relaxation periods or high-precision measurements, and their sensitivity tends to diminish with aging or variations in state of health [20,21]. Voltage relaxation profiling and coulombic efficiency monitoring have also been employed to infer reversible or irreversible plating behavior, although these approaches require stable testing environments and often fail to provide real-time information [22,23]. Recently, emerging techniques such as dynamic capacitance measurement (DCM) [24] and data-driven or machine-learning-assisted models [25,26] have demonstrated potential for rapid, non-invasive, and online detection of lithium plating. Nevertheless, developing a detection methodology that is simultaneously simple, low-cost, and compatible with real-time battery management system (BMS) integration remains a major challenge for the practical deployment of fast-charging lithium-ion batteries.

Commercial lithium-ion batteries typically employ a constant-current–constant-voltage (CC–CV) or multi-step CC–CV fast-charging protocol [27]. The transient kinetics that occur at the moment of transition from the constant-current (CC) to the constant-voltage (CV) stage contain rich information about the interfacial state of the electrodes. At the end of the CC stage, the anode potential approaches the lithium plating potential (0 V vs. Li^+/Li). The occurrence of lithium deposition in any localized region alters the surface properties and charge-transfer

characteristics of the anode, which should be clearly reflected in the current decay behavior immediately following the CC–CV transition. However, this specific transient signal has been largely overlooked as a diagnostic indicator of lithium plating.

In this work, we propose a lithium plating detection approach based on the analysis of transient current kinetics during the CC–CV switching process. The main contributions of this study are threefold. First, we introduce a simple and computationally efficient lithium plating indicator derived from the instantaneous current response within the initial stage of the CV phase, enabling rapid and non-invasive detection without the need for additional hardware or complex signal injection. Second, a physics-based electrochemical model incorporating lithium plating side reactions is employed to provide a mechanistic interpretation of the proposed indicator and to systematically evaluate its sensitivity and robustness under various operating conditions, with benchmarking against conventional relaxation-voltage-based detection methods. Third, the proposed approach is validated using publicly available datasets, demonstrating its robustness and general applicability under realistic charging scenarios. Overall, the results of this study provide a physically grounded and implementation-friendly diagnostic pathway for enhancing lithium plating awareness in practical BMS applications.

2. Method

As illustrated in Fig. 1, the proposed lithium plating detection approach exploits the transient current response at the CC–CV transition during charging. Fig. 1a illustrates the CC–CV charging protocol and the extraction of the short-time current feature parameter γ at the CC–CV transition, while Fig. 1b presents the evolution of the extracted feature during long-term cycling and its application to lithium plating diagnosis.

To elucidate the physical origin of the transient current response during the transition from CC to CV charging, a mechanistic electrochemical model was developed based on the Doyle–Fuller–Newman

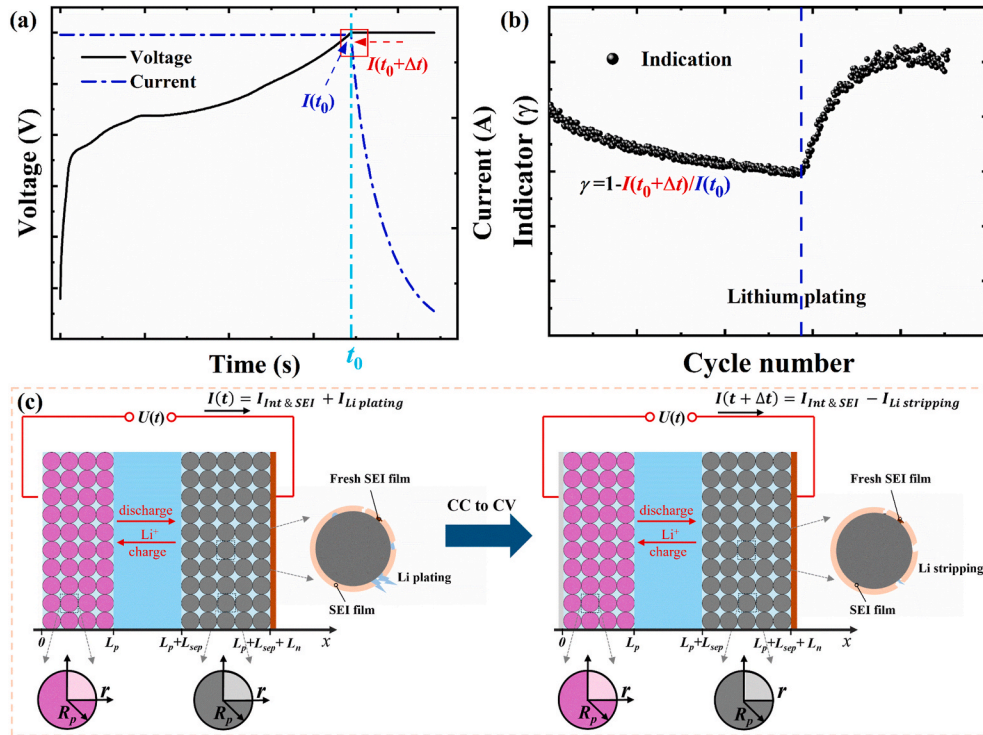


Fig. 1. Overview of the proposed lithium plating detection framework. (a) CC–CV charging protocol and identification of the transition point. (b) Extraction of the transient current feature γ within the first Δt seconds of the CV stage. (c) Mechanistic interpretation based on a physics-based electrochemical model incorporating lithium plating.

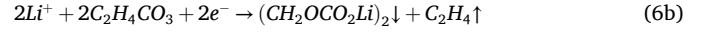
pseudo-two-dimensional (DFN-P2D) model framework (Fig. 1c). The model simultaneously accounts for lithium intercalation, solid electrolyte interphase (SEI) formation, and lithium plating/stripping reactions at the negative electrode, thereby enabling the interpretation of current transients in terms of underlying interfacial kinetics.

2.1. DFN model of Li-ion battery

The Doyle–Fuller–Newman (DFN) framework is extensively employed to simulate the electrochemical behaviour of lithium-ion batteries [28–31]. As illustrated in Fig. 1a, the cell configuration includes two current collectors, a positive and a negative electrode, and a separator positioned between them. Each electrode is treated as a porous medium composed of spherical active material particles. The DFN formulation incorporates two primary spatial coordinates: one along the cell thickness, representing transport processes in the electrolyte phase, and another along the particle radius, describing lithium diffusion within the solid phase. These dimensions collectively capture ion migration, concentration gradients, and solid-state diffusion phenomena. As summarized in Table 1 [28], the model is governed by coupled equations describing charge conservation in both phases, species conservation in the electrolyte and solid, and interfacial reaction kinetics.

2.2. Modeling of side reaction

In this work, both SEI formation and lithium plating are incorporated as parasitic reactions at the graphite anode, in addition to the main lithium intercalation process. Accordingly, three electrochemical reactions occur at the anode surface (Eq. (6)): (a) lithium intercalation into graphite, (b) electrolyte reduction leading to SEI formation, and (c) metallic lithium deposition.



To account for heterogeneous surface transport, a dual-path model is employed [32], consisting of crack regions and dense SEI regions formed during graphite expansion (Fig. 1). The total interfacial current density is expressed as the sum of the three reactions occurring over both regions:

$$i_{tot} = i_{cov} + i_{crd} = \sum (i_{int} + i_{SEI} + i_{pl})_{cov,crd} \quad (7)$$

where subscripts cov and crd denote the dense SEI-covered and cracked areas, respectively.

For the SEI-covered region, the intercalation current density, $i_{int,cov}$, is defined as:

$$i_{int,cov} = \varepsilon_{cov} i_{int}^0 \left(\exp \left(\frac{\alpha_{a,int} F}{RT} \eta_{int,cov} \right) - \exp \left(- \frac{\alpha_{c,int} F}{RT} \eta_{int,cov} \right) \right) \quad (8)$$

$$\eta_{int,cov} = \phi_s - \phi_e - U_{int} - i_{cov} R_{film} \quad (9)$$

In the cracked region, direct contact between graphite and electrolyte is assumed, and SEI resistance is neglected:

$$i_{int,crd} = \varepsilon_{crd} i_{int}^0 \left(\exp \left(\frac{\alpha_{a,int} F}{RT} \eta_{int,crd} \right) - \exp \left(- \frac{\alpha_{c,int} F}{RT} \eta_{int,crd} \right) \right) \quad (10)$$

$$\eta_{int,crd} = \phi_s - \phi_e - U_{int} \quad (11)$$

Table 1

Equations of DFN model used in the study [28].

Description	Equation	Boundary condition
Electrode		
Mass conservation	$\frac{\partial c_s}{\partial t} = \frac{1}{r^2} \frac{\partial}{\partial r} \left(D_s r^2 \frac{\partial c_s}{\partial r} \right) \quad (1)$	$\begin{cases} \left. \frac{\partial c_s(r, t)}{\partial r} \right _{r=0} = 0 \\ \left. \frac{\partial c_s(r, t)}{\partial r} \right _{r=R_p} = \frac{i_{loc}}{D_s F} \end{cases}$
Charge conservation	$\nabla \cdot (\sigma_s^{\text{eff}} \nabla \phi_s) = i_{v,tot} \quad (2)$	$\begin{cases} \sigma_{s,n}^{\text{eff}} \frac{\partial \phi_s(x, t)}{\partial x} \Big _{x=0} = \sigma_{s,p}^{\text{eff}} \frac{\partial \phi_s(x, t)}{\partial x} \Big _{x=L_n+L_{sep}+L_p} = \pm \frac{I}{A} \\ \left. \frac{\partial \phi_s(x, t)}{\partial x} \right _{x=L_n} = \left. \frac{\partial \phi_s(x, t)}{\partial x} \right _{x=L_n+L_p} = 0 \end{cases}$
	$\sigma_s^{\text{eff}} = \varepsilon_s^{\text{brugg}} \sigma_s$	
Electrolyte		
Mass conservation	$\varepsilon_e \frac{\partial c_e}{\partial t} = \nabla \cdot (D_e^{\text{eff}} \nabla c_e) + (1 - t_+^0) \frac{i_{v,tot}}{F} \quad (3)$	$\begin{cases} D_{e,n}^{\text{eff}} \frac{\partial c_e(x, t)}{\partial x} \Big _{x=0} = D_{e,p}^{\text{eff}} \frac{\partial c_e(x, t)}{\partial x} \Big _{x=L_n+L_{sep}+L_p} = 0 \\ D_{e,n}^{\text{eff}} \frac{\partial c_e(x, t)}{\partial x} \Big _{x=L_n} = D_{e,sep}^{\text{eff}} \frac{\partial c_e(x, t)}{\partial x} \Big _{x=L_n^+} \\ D_{e,sep}^{\text{eff}} \frac{\partial c_e(x, t)}{\partial x} \Big _{x=(L_n+L_{sep})^-} = D_{e,p}^{\text{eff}} \frac{\partial c_e(x, t)}{\partial x} \Big _{x=(L_n+L_{sep})^+} \end{cases} \quad (4)$
Charge conservation	$\begin{aligned} D_e^{\text{eff}} &= \varepsilon_e^{\text{brugg}} D_e \\ \nabla \cdot \left(\kappa_e^{\text{eff}} \nabla \phi_e - \frac{2\kappa_e^{\text{eff}} RT}{F} \left(1 + \frac{d \ln f}{d \ln c_e} \right) (1 - t_+^0) \nabla \ln c_e \right) &= -i_{v,tot} \\ \kappa_e^{\text{eff}} &= \varepsilon_e^{\text{brugg}} \kappa_e \end{aligned}$	$\begin{cases} \phi_e(x, t) \Big _{x=0} = \phi_e(x, t) \Big _{x=L_n+L_{sep}+L_p} = 0 \\ \kappa_{e,n}^{\text{eff}} \frac{\partial \phi_e(x, t)}{\partial x} \Big _{x=L_n} = \kappa_{e,sep}^{\text{eff}} \frac{\partial \phi_e(x, t)}{\partial x} \Big _{x=L_n^+} \\ \kappa_{e,sep}^{\text{eff}} \frac{\partial \phi_e(x, t)}{\partial x} \Big _{x=(L_n+L_{sep})^-} = \kappa_{e,p}^{\text{eff}} \frac{\partial \phi_e(x, t)}{\partial x} \Big _{x=(L_n+L_{sep})^+} \end{cases} \quad (5)$
Reaction kinetics		
Butler-Volmer	$i_{int} = i_{int}^0 \left(\exp \left(\frac{\alpha_{a,int} F}{RT} \eta_{int} \right) - \exp \left(- \frac{\alpha_{c,int} F}{RT} \eta_{int} \right) \right)$	
Exchange current	$i_{int}^0 = F k_{int}^0 c_{s,max}^{\alpha_a} (c_{s,max} - c_{s,surf})^{\alpha_a} c_{s,surf}^{\alpha_c}$	
Overpotential	$\eta_{int} = \phi_s - \phi_e - U_{int} - i_{loc} R_{film}$	
	$i_{v,tot} = \sum i_v = \sum \alpha_s i_{loc} = \alpha_s i_{int}$	

The crack ratio ε_{crd} evolves with graphite expansion as:

$$\varepsilon_{crd} = a_{crd} \left((1 + \Delta V)^{\frac{2}{3}} - 1 \right) \quad (12)$$

Here, a_{crd} represents a scaling factor for the volumetric expansion ΔV , as illustrated in Fig. S1. The dependence of ΔV on the SOC of negative electrode is evaluated through both experimental measurements and theoretical modeling [33].

2.2.1. SEI formation

The SEI formation is modeled as an irreversible cathodic reaction,

$$i_{lpl,crd} = \begin{cases} \varepsilon_{crd} i_{lpl}^0 \left(\exp\left(\frac{\alpha_{a,lpl} F}{RT} \eta_{lpl,crd}\right) - \exp\left(-\frac{\alpha_{c,lpl} F}{RT} \eta_{lpl,crd}\right) \right), & \eta_{lpl,crd} < 0 \\ \varepsilon_{crd} i_{lpl}^0 \left(\exp\left(\frac{\alpha_{a,lpl} F}{RT} \eta_{lpl,crd}\right) - \exp\left(-\frac{\alpha_{c,lpl} F}{RT} \eta_{lpl,crd}\right) \right) \frac{\beta n_{lpl,crd}}{1 + \beta n_{lpl,crd}}, & \eta_{lpl,crd} \geq 0 \end{cases}$$

governed by the reduction of ethylene carbonate (EC). Just like Eq. (8) (9), the current density $i_{SEI,cov}$ in the SEI coverage area is defined as [34]:

$$i_{SEI,cov} = -\varepsilon_{cov} F k_{SEI}^0 c_{EC,surf}^2 \exp\left(-\frac{\alpha_{c,SEI} F}{RT} \eta_{SEI,cov}\right) \quad (13)$$

$$\eta_{SEI,cov} = \phi_s - \phi_e - U_{SEI} - i_{cov} R_{film} \quad (14)$$

where k_{SEI}^0 is the kinetic rate constant, $\alpha_{c,SEI}$ is the cathodic charge transfer coefficient, U_{SEI} is the equilibrium potential of SEI formation reaction. And $i_{SEI,cov}$ is defined as:

$$i_{SEI,crd} = -\varepsilon_{crd} F k_{SEI}^0 c_{EC,surf}^2 \exp\left(-\frac{\alpha_{c,SEI} F}{RT} \eta_{SEI,crd}\right) \quad (15)$$

$$\eta_{SEI,crd} = \phi_s - \phi_e - U_{SEI} \quad (16)$$

The surface EC concentration $c_{EC,surf}$ is determined by the steady-state balance between diffusion and consumption:

$$D_{EC} \frac{c_{EC,surf} - c_{EC}^0}{\delta_{film}} = -\frac{i_{SEI}}{F} \quad (17)$$

where D_{EC} is the diffusivity of EC, c_{EC}^0 is the concentration of EC in the bulk electrolyte, and δ_{film} is the thickness of the surface film.

2.2.2. Lithium plating/stripping

The side reaction of lithium plating/stripping can be described by the ButlerVolmer equation [35], $i_{lpl,cov}$ is defined as:

$$i_{lpl,cov} = \begin{cases} \varepsilon_{cov} i_{lpl}^0 \left(\exp\left(\frac{\alpha_{a,lpl} F}{RT} \eta_{lpl,cov}\right) - \exp\left(-\frac{\alpha_{c,lpl} F}{RT} \eta_{lpl,cov}\right) \right), & \eta_{lpl,cov} < 0 \\ \varepsilon_{cov} i_{lpl}^0 \left(\exp\left(\frac{\alpha_{a,lpl} F}{RT} \eta_{lpl,cov}\right) - \exp\left(-\frac{\alpha_{c,lpl} F}{RT} \eta_{lpl,cov}\right) \right) \frac{\beta n_{lpl,cov}}{1 + \beta n_{lpl,cov}}, & \eta_{lpl,cov} \geq 0 \end{cases}$$

$$\eta_{lpl,cov} = \phi_s - \phi_e - U_{lpl} - i_{cov} R_{film} \quad (19)$$

$$i_{lpl}^0 = F k_{lpl}^0 c_e^{\alpha_{a,lpl}} \quad (20)$$

where Eq. (18a) and Eq. (18b) describe lithium plating and stripping, respectively. i_{lpl}^0 is the exchange current density of lithium plating, $\alpha_{a,lpl}$ and $\alpha_{c,lpl}$ respectively denote the electroplating reaction anodic and cathodic charge transfer coefficient, $\eta_{lpl,cov}$ is the electroplating reaction overpotential. $\frac{\beta n_{lpl,cov}}{1 + \beta n_{lpl,cov}}$ is the correction term considering the limitation by the amount of reversible lithium $n_{lpl,cov}$. β is a large value to effectively make the term $\frac{\beta n_{lpl,cov}}{1 + \beta n_{lpl,cov}}$ equal to 1 when $n_{lpl,cov} \gg 1$, and 0 when $n_{lpl,cov} = 0$, as in Ref. [35]. And it is assumed that the amount of dead lithium is determined by $n_{dead,cov} = \theta n_{lpl,cov}$, where θ is the molar fraction. Similarly, $i_{lpl,crd}$ is defined as follows:

$$\eta_{lpl,crd} = \phi_s - \phi_e - U_{lpl} \quad (22)$$

Within the dual-path model framework, the crack ratio determines how different electrochemical reactions are distributed between the crack (crd) and SEI-covered (cov) regions, as illustrated in Fig. S2. For lithium intercalation and SEI formation, the crack ratio mainly governs the partitioning of reaction rates between these two regions, with both reactions occurring in crack and covered areas. In contrast, lithium plating exhibits a preferential spatial behavior. In the early stage, plating predominantly occurs in the crack regions, where the SEI layer is thinner or partially ruptured. As degradation progresses, lithium plating may gradually extend to the SEI-covered regions under more severe conditions.

2.2.3. Composite film growth and resistance

During battery operation, a composite interfacial layer consisting of the SEI and plated metallic lithium gradually develops on the anode surface. The total thickness of this layer, denoted as δ_{film} , represents the sum of three components: the SEI thickness (δ_{SEI}) and the lithium deposition thickness (δ_{lpl}), and the initial interfacial layer thickness ($\delta_{film,0}$). This relationship can be written as:

$$\delta_{film} = \delta_{SEI} + \delta_{lpl} + \delta_{film,0} = \frac{c_{SEI} M_{SEI}}{\alpha_s \rho_{SEI}} + \frac{c_{lpl} M_{lpl}}{\alpha_s \rho_{lpl}} + \delta_{film,0} \quad (23)$$

Here c_{SEI} and c_{lpl} denote the molar concentrations of SEI species and deposited lithium per unit electrode volume, respectively. Their tem-

poral evolution is governed by:

$$\frac{\partial c_{SEI}}{\partial t} = -\frac{\alpha_s i_{SEI}}{2F} \quad (24)$$

$$\frac{\partial c_{lpl}}{\partial t} = -\frac{\alpha_s i_{lpl}}{F} \quad (25)$$

In the present formulation, only the regions covered by the composite layer are considered to contribute to the effective film growth, while cracked or uncovered areas are assumed inactive in thickness accumulation. Therefore, $c_{SEI} = c_{SEI,cov}$, and $c_{lpl} = c_{lpl,cov}$. The parameters ρ_{SEI} and ρ_{lpl} refer to the densities of the SEI and metallic lithium, respectively, whereas M_{SEI} and M_{lpl} correspond to their molar masses.

The ionic resistance across the composite film R_{film} is defined by:

$$R_{film} = \frac{\delta_{film}}{\kappa_{film}} \quad (26)$$

where κ_{film} represents the effective ionic conductivity of the interfacial layer. Furthermore, as the composite film thickens, the volume fraction of the electrolyte phase (ε_e) decreases according to Refs. [34,35]:

$$\frac{\partial \varepsilon_e}{\partial t} = -\alpha_s \frac{\partial \delta_{film}}{\partial t} \quad (27)$$

2.3. Lumped thermal model

The change of battery temperature follows the energy balance equation:

$$\rho_{bat} C_p \frac{\partial T}{\partial t} = Q + h a_{bat} (T_{amb} - T) \quad (28)$$

where ρ_{bat} is the density of the cell, C_p is the effective specific heat capacity, and T is the cell temperature. Q represents the total heat generation within the battery [35], h is the effective heat transfer coefficient, a_{bat} is the specific surface area of the battery, and T_{amb} is the ambient temperature.

Some parameters in the electrochemical model are temperature-dependent [35], such as the solid-phase diffusion coefficient and reaction rate constant. Their values at a given temperature T can be described by an Arrhenius-type relationship:

$$k(T) = k_{ref} \cdot \exp \left[\frac{E_{a,k}}{R} \left(\frac{1}{T_{ref}} - \frac{1}{T} \right) \right] \quad (29)$$

Where $k(T)$ represents the temperature-dependent parameter, k_{ref} is the reference value at temperature T_{ref} , $E_{a,k}$ is the activation energy.

2.4. Simulation of CC–CV transition

To capture the transient behavior at the CC–CV switching point, the model was operated under a galvanostatic charging condition until the cell voltage reached the preset cutoff U_{CV} , at which point the control mode switched to potentiostatic operation. The instantaneous current decay during the first 10 s after the transition was numerically resolved using a time step of 1 ms to ensure adequate temporal accuracy.

2.5. Definition of the detection indicator

During the transition from CC to CV charging, the applied current exhibits a rapid decay immediately after the voltage reaches the cut-off value. This transient response reflects the dynamic redistribution of lithium ions near the anode surface, as well as the activation of side reactions such as lithium plating. To quantify this short-term current behavior, a dimensionless indicator, denoted as γ is introduced to characterize the relative current decay rate within a short time window following the CC–CV transition.

Let t_0 denote the switching time from CC to CV charging, and $I(t)$ the cell current at time t . It should be noted that the current $I(t_0)$ at time t_0 is exactly the current during the constant current phase. The indicator γ is defined as:

$$\gamma = \frac{I(t_0) - I(t_0 + \Delta t)}{I(t_0)} \quad (30)$$

where Δt represents the observation window after the transition, typically chosen as 10 s in this study.

2.6. Quantitative detection criterion for plating onset

To enable objective and automated identification of the onset of lithium plating from the cycle-wise evolution of γ , a sliding-window detection criterion was introduced. For each cycle index N , a local window W_N containing the most recent w is defined as:

$$W_N = \{\gamma(N-w+1), \gamma(N-w+2), \dots, \gamma(N)\} \quad (31)$$

Within each window, the local trend of γ was quantified using linear regression, and the fitted slope was denoted as a_N . Under plating-free conditions, γ typically shows a decreasing or nearly stable trend, corresponding to $a_N \leq 0$. Once plating emerges, γ exhibits a persistent increasing trend, and thus a_N becomes positive.

To avoid false triggering caused by measurement noise or random fluctuations, the onset of lithium plating is identified only when the following two conditions are satisfied simultaneously.

- (1) $a_N > \delta_a$, where δ_a is a small positive threshold;
- (2) The condition $a_N > \delta_a$ persists for M consecutive windows.

Here, δ_a and M are hyperparameters that can be selected based on the noise level of the dataset and the desired robustness. This criterion enables automatic, reproducible, and objective detection of the transition point in γ without relying on subjective visual inspection.

3. Results and discussion

3.1. Model validation

The developed model was validated against open-access experimental data reported by CALCE Battery Research Group [36]. As shown in Fig. 3a, the simulated voltage profiles exhibit good agreement with the measured results across various C-rates, with a root-mean-square error below 1.5%. The model also reproduces the gradual capacity fade observed during prolonged cycling, confirming its ability to capture the essential degradation mechanisms. Under fast-charging protocols, the model is also able to match the experimental results [37], as shown in Fig. 2c.

The chemistries of the experimental datasets are explicitly clarified here. The CALCE dataset [36] corresponds to LiCoO₂/graphite cells, whereas the fast-charging dataset reported by Attia et al. [37] involves LiFePO₄/graphite chemistry. The model developed in this work is primarily parameterized for LiCoO₂/graphite systems, with the corresponding parameters listed in Table S1 and S2, and all subsequent simulations are conducted based on this chemistry. In contrast, the LiFePO₄/graphite dataset is employed solely to validate the generality and robustness of the governing equations under different chemistries. For this purpose, the model parameters are adopted from Liu et al. [38]. Therefore, the results (Fig. 2c) associated with LiFePO₄/graphite are intended only for model verification, rather than for quantitative prediction or further analysis.

3.2. Indicator behavior under plating-free and plating conditions

Fig. 3a presents the simulated capacity retention at a cycling rate of 0.5C for models with and without lithium plating. It can be observed that during the early cycling stage (prior to the 50th cycle), both cases exhibit a rapid capacity decay, irrespective of whether lithium plating is included. This behavior arises because, at the initial stage of cycling,

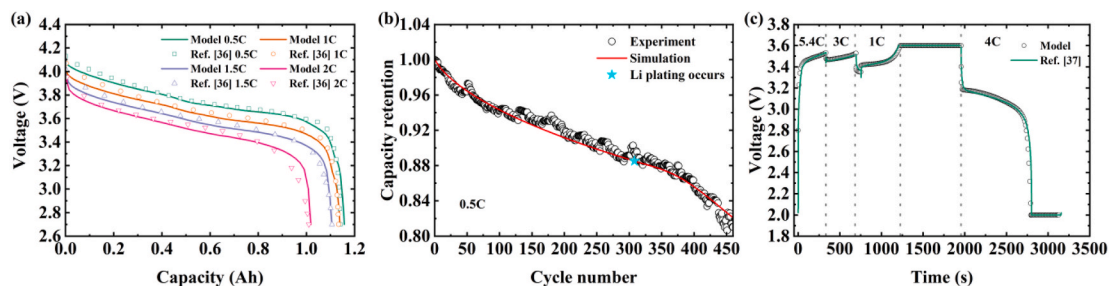


Fig. 2. Comparison of model results with experiment data in terms of (a) discharge voltage profiles at rates of 0.5C, 1C, 1.5C, and 2C; (b) the evolution of discharge capacity with cycle count during 0.5C long-term cycling; (c) voltage profiles at fast-charging protocols.

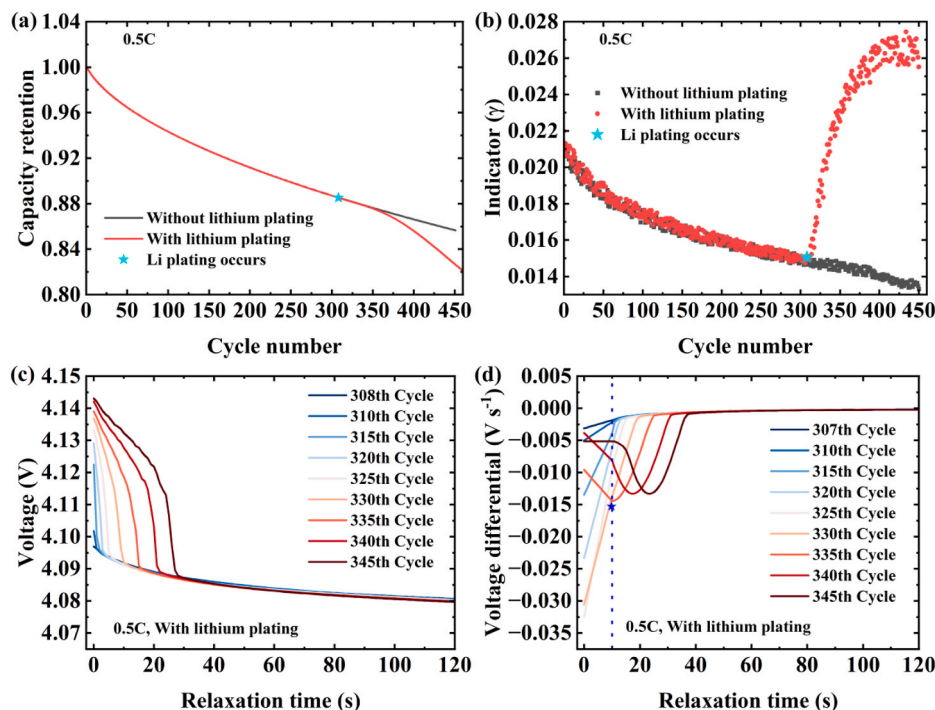


Fig. 3. Comparison of capacity fade, indicator evolution, and voltage-relaxation signatures under plating-free and plating conditions. (a) Simulated capacity retention at 0.5C cycling for models with and without lithium plating. (b) Evolution of the proposed transient-current indicator γ for plating-free and plating cases. (c) Voltage relaxation profiles following CC charging at selected cycles (from the 308th to the 345th cycle) for the model with lithium plating. (d) Corresponding time derivative of the relaxation voltage ($\frac{dV}{dt}$) for the same cycles, highlighting the weak and delayed stripping-induced features.

capacity degradation is dominated by parasitic side reactions associated with SEI formation. During this period, the SEI layer forms rapidly on the anode surface, and as the SEI thickens (Fig. S3), solvent diffusion through the film becomes increasingly hindered, as described by Eq. (17). Consequently, the SEI growth rate gradually decreases and eventually approaches a quasi-steady state. As a result, during the intermediate cycling stage (up to the 308th cycle), the capacity retention exhibits an approximately linear decline with cycle number.

For the model incorporating lithium plating, lithium deposition initiates at the end of the CC charging stage in the 308th cycle, as confirmed by the lithium plating reaction rate shown in Fig. S4b. In contrast, for the model without lithium plating, the capacity continues to decrease linearly throughout the entire cycling process. Once lithium plating occurs, the plating-enabled model exhibits a markedly accelerated capacity fade, and a clear divergence in discharge capacity between the two models emerges at approximately the 350th cycle.

Throughout the cycling process, the evolution of the proposed indicator γ , extracted according to Eq. (30), is shown in Fig. 3b. Under plating-free conditions, γ evolves consistently with the trend of capacity retention, displaying a gradual monotonic decrease with cycling. In

contrast, when lithium plating is present, γ exhibits an abrupt transition at the cycle where plating initiates (308th cycle), changing from a decreasing trend to a pronounced increase. Notably, γ continues to increase over the subsequent 100 cycles. This distinct behavior provides a clear diagnostic signature of lithium plating. Compared with the sudden drop in capacity retention, the proposed indicator enables the detection of lithium plating approximately 50 cycles earlier. These results demonstrate that the proposed transient-current indicator effectively discriminates between plating-free and plating conditions and provides an early-warning signal for the onset of lithium plating.

To further evaluate the effectiveness of the proposed method, a comparison was conducted between the transient-current indicator γ and the conventional voltage relaxation method (shown in Fig. 3c and d). The latter typically relies on identifying a voltage plateau or a characteristic peak in the $\frac{dV}{dt}$ curve during the rest period following a charge stroke, which requires a significant duration of inactivity to ensure diagnostic accuracy. In contrast, the proposed indicator γ is extracted directly from the transient current response during the operational phase, eliminating the need for extended rest periods. Furthermore, while voltage relaxation methods often require a substantial

amount of accumulated lithium plating to produce a detectable signal, the indicator γ exhibits a sensitive 'inflection point' at the very onset of plating (the 308th cycle). As shown in Fig. 3d, compared to the voltage relaxation method, this provides a diagnostic lead time of approximately 30 cycles before a clear peak appears in the relaxation voltage signal. This suggests that the kinetic-based indicator γ offers superior sensitivity and real-time monitoring capabilities over the traditional thermodynamic-based relaxation analysis.

3.3. Mechanistic interpretation

To elucidate the physical origin of the proposed transient-current indicator γ , the evolution of reaction current partitioning and interfacial kinetics during the transition from constant-current (CC) to constant-voltage (CV) charging was analyzed using the mechanistic model and equivalent-circuit.

During the CC charging stage, the applied current is externally imposed and must be sustained regardless of the internal electrochemical state of the cell. At the negative electrode, the total applied current can be decomposed into the graphite intercalation current, SEI formation current and the lithium plating current, such that

$$I_{app} = I_{int\&SEI} + I_{pl} \quad (32)$$

where $I_{int\&SEI}$ represents the lithium intercalation and SEI formation reaction current, I_{pl} denotes the lithium plating reaction current. Owing to the galvanostatic constraint, the applied current I_{app} remains fixed by the charging protocol. When the lithium plating reaction occurs, the increase in I_{pl} leads to a corresponding decrease in $I_{int\&SEI}$, as shown in Fig. 4a.

Upon switching from CC to CV charging, the control variable changes from current to voltage. Consequently, the total charging current is no longer prescribed but instead determined by the instantaneous electrochemical impedance of the cell. From an equivalent-circuit perspective, the applied current during the CV stage can be expressed as

$$I_{app} = \frac{U}{R_s + \left(\frac{1}{\frac{1}{R_{int}} + \frac{1}{R_{pl}}} \right)} \quad (33)$$

where U is the applied terminal voltage, R_s denotes the effective ohmic resistance, and R_{int} and R_{pl} represent the effective resistances associated with the intercalation and plating reactions at the negative electrode, respectively. It should be emphasized that R_s , R_{int} , and R_{pl} are introduced here as effective lumped quantities to facilitate physical interpretation of the current redistribution at the CC-CV transition. They are not intended to represent uniquely identifiable circuit elements independently fitted from experimental impedance data. Instead, this equivalent-circuit description serves as a conceptual bridge between the DFN model and the observed transient current response.

At the onset of CV charging, the electrode polarization rapidly decreases as the cell voltage is held constant (shown in Fig. S5). The reduction in overpotential suppresses the lithium plating reaction, causing the corresponding current to diminish sharply or even reverse, transitioning into lithium stripping and giving rise to a reverse current, as shown in Fig. 4b. In terms of the equivalent circuit, this behavior is reflected by a rapid increase in R_{pl} , which may approach infinity when plating is completely inhibited. The sudden removal or weakening of the plating branch increases the overall equivalent resistance, leading to an abrupt decrease in the total applied current. When the reaction transitions to lithium stripping, the total applied current is further reduced, and Eq. (33) transforms into the following expression.

$$I_{app} = \frac{U}{R_s + R_{int}} - I_{stripping} \quad (34)$$

where $I_{stripping}$ represents lithium stripping current. Furthermore, Fig. 4c presents a conceptual equivalent circuit schematic that qualitatively illustrates how variations in interfacial resistance may give rise to the observed current transient, thereby providing an intuitive electrochemical interpretation of the proposed indicator γ .

It is noteworthy that the intercalation-related current $I_{int\&SEI}$ does not recover to an independent reference level at the onset of CV charging, but instead continues from its value at the end of the CC stage. As a result, when lithium plating occurs during CC charging, the suppression of intercalation caused by the diversion of current to the plating reaction is naturally carried over to the CV stage. Consequently, the interfacial reaction state at the CC-CV transition reflects not only the instantaneous suppression of lithium plating induced by voltage control, but also the cumulative influence of plating during the CC stage on the intercalation

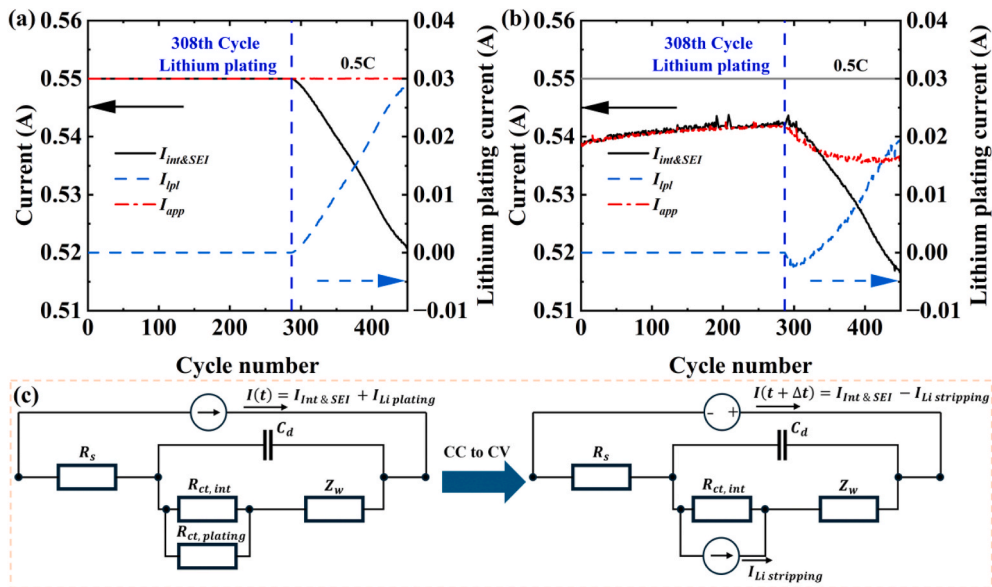


Fig. 4. Evolution of current distribution and mechanistic interpretation of the transient-current feature at the CC-CV transition. (a) Current distribution at the end of CC charging for different cycle numbers. (b) Current distribution after 10 s of CV charging for different cycle numbers. (c) Simplified equivalent circuit schematic for visualizing the effect of interfacial resistance variations on the transient current feature.

pathway.

Overall, the transient-current indicator γ arises from the coupled effects of reaction-current inheritance and abrupt kinetic redistribution during the CC–CV transition. When lithium plating occurs during the CC stage, part of the applied current is diverted from intercalation and retained as a suppressed intercalation current at the onset of CV charging. Upon switching to voltage control, the rapid reduction in polarization strongly suppresses the plating reaction, leading to a sudden increase in the effective plating resistance and a sharp decrease (or reversal) of the associated reaction current. The superposition of the reduced intercalation current inherited from the CC stage and the abrupt collapse of the plating current during early CV charging gives rise to a pronounced current transient. This distinct and physically well-defined response forms the mechanistic basis of γ , enabling robust identification of lithium plating through short-time current dynamics at the CC–CV transition.

3.4. Robustness under different C-rates and multi-CC–CV protocols

To evaluate the generality of the proposed indicator γ , simulations were performed under a range of charging rates and multi-step CC–CV protocols. Five charging rates—0.2C, 0.5C, 0.6C, 0.8C, and 1.5C—were selected to assess the effectiveness of the indicator under different operating conditions. The corresponding evolutions of capacity retention and γ with cycle number are shown in Fig. 5a–c and Fig. S6a–b, respectively. The temperature variations during charging at different selected rates are presented in Fig. S7a.

For all charging rates except 1.5C, γ exhibits a consistent evolution pattern: it decreases monotonically during the early cycling stage and increases sharply once lithium plating occurs. In particular, at 0.5C and 0.6C, γ increases rapidly after the onset of lithium plating and subsequently approaches a quasi-steady value as cycling proceeds. At 1.5C, lithium plating is predicted from the first cycle, and γ therefore remains at a persistently elevated level without exhibiting a distinct trend-reversal onset. This result highlights an important limitation of the method: γ is most informative as an onset detector when a plating-free baseline exists, whereas under immediate-plating conditions it should instead be interpreted as a persistent plating-risk indicator. Based on the evolution of γ , the initial cycles associated with lithium plating onset at 0.2C, 0.5C, 0.6C, 0.8C, and 1.5C are identified as the 1389th, 308th, 147th, 26th, and 1st cycles, respectively. These results are further corroborated by the cycle-dependent evolution of the lithium plating reaction rate at the negative electrode surface, as shown in Fig. S4. Additionally, under a low-temperature condition of 10 °C at 0.2C, γ exhibits a similar evolution behavior, with a pronounced increase at the onset of lithium plating, as shown in Fig. S8. Collectively, these findings indicate that the proposed indicator is highly robust and capable of detecting the onset of lithium plating across a wide range of charging rates.

To further examine the applicability of γ under more complex charging conditions, two types of multi-step CC–CV charging protocols were designed following [37], with detailed charging steps summarized in Table S3 and S4. Protocol Type 1 consists of three consecutive CC–CV

charging steps, allowing three indicators to be extracted according to Eq. (30). In this study, particular attention is paid to the latter two CC–CV transitions, where lithium plating is more likely to occur. Fig. 6a and b presents the simulation results for the Type 1-1 and Type 1-2 protocols, respectively. The corresponding temperature evolution during the charging stages for both protocols is shown in Fig. S7b. The bottom panels of Fig. 6a and b shows the evolution of the lithium plating reaction rate at the negative electrode surface with cycle number, from which the initial plating cycles within the latter two CC–CV stages are identified as the 148th and 162nd cycles for Type 1-1, and the 59th and 190th cycles for Type 1-2. Correspondingly, the evolution of γ exhibits clear inflection points at the same cycle numbers. In contrast, these early transitions cannot be reliably identified from the capacity retention curves alone, demonstrating the superior sensitivity of the proposed indicator under complex charging protocols.

Similar to Type 1, Protocol Type 2 consists of multiple CC steps followed by a single CV step, resulting in one CC–CV transition from which a single indicator can be extracted using Eq. (30). Fig. 7a and b shows the simulation results for the Type 2-1 and Type 2-2 protocols, respectively. The lithium plating reaction rate indicates that lithium plating initiates during different CC steps for each protocol, while the proposed indicator γ corresponds specifically to the CC–CV transition at a charging rate of 0.5C. Accordingly, the initial lithium plating within the last CC stage cycles detected by γ are the 318th cycle for Type 2-1 and the 154th cycle for Type 2-2. Notably, γ accurately identifies these transition points even when lithium plating has already occurred during preceding CC steps.

Overall, these results demonstrate that γ is robust with respect to both charging-rate variations and CC–CV protocol design, confirming that it reflects a fundamental electrochemical transition rather than a protocol-specific artifact.

3.5. Validation with open-source experimental datasets

To further assess the practical applicability of the proposed transient-current indicator γ , its performance was validated using independent open-source experimental datasets. Two publicly available datasets were considered: (i) the CALCE lithium-ion battery dataset and (ii) a fast-charging lithium-ion battery dataset reported in Ref. [37].

Both datasets comprise long-term cycling experiments conducted under CC–CV charging protocols and provide high-resolution current and voltage measurements around the CC–CV transition. These features make them well suited for evaluating the proposed indicator, which is extracted from the short-time current transient immediately following the CC–CV switching point. For each cycle, γ was calculated using the same definition and extraction procedure as employed in the model-based analysis, ensuring methodological consistency between simulation and experimental validation. For clarity of visualization, the blue solid lines in Fig. 8 and Fig. S9 represent the indicator γ after Savitzky–Golay (SG) smoothing, and the blue dashed lines denote the detected onset of lithium plating identified using the quantitative detection criterion proposed in Section 2.6. The SG filter parameters used in this work were a window length of 11 cycles and a polynomial order of 2.

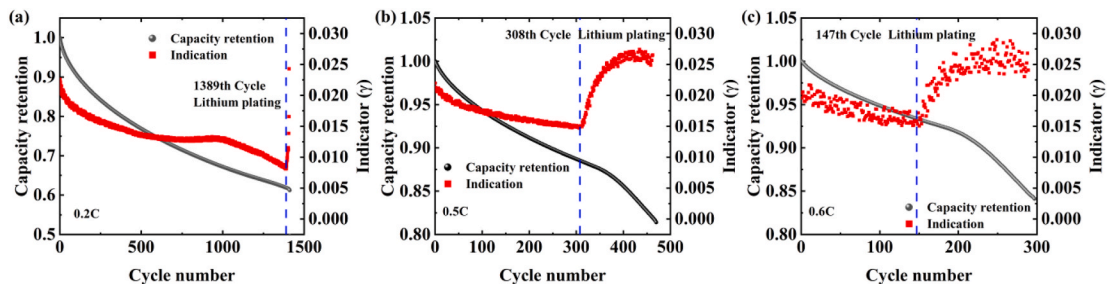


Fig. 5. Robustness of the transient-current indicator γ under (a–c) three different charging rates, 0.2C, 0.5C, and 0.6C.

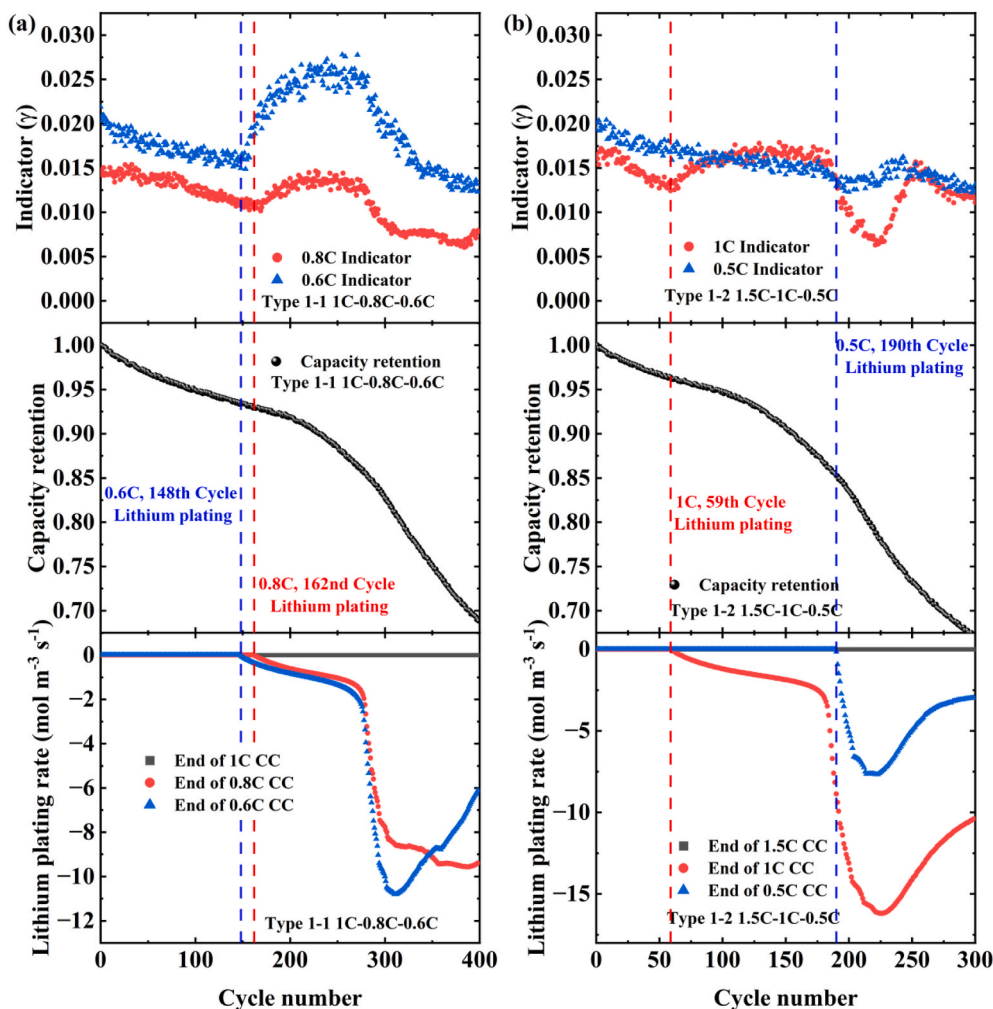


Fig. 6. Simulation results of indicator behavior for two multi-step CC-CV charging protocols (Type 1). The left and right sets of panels correspond to the Type 1-1 (a) and Type 1-2 (b) protocols, with the top, middle, and bottom panels showing the capacity retention, the evolution of γ , and the lithium plating reaction rate, respectively.

Fig. 8a and b presents the evolution of γ together with the corresponding capacity retention as a function of cycle number for representative CALCE cells. During the early and intermediate stages of cycling, γ decreases gradually, consistent with the smooth aging behavior observed in the discharge capacity. With continued cycling, γ exhibits a distinct trend transition (i.e., at the 417th cycle in Fig. 8a and the 289th cycle in Fig. 8b), changing from a decreasing to an increasing evolution. Notably, this transition occurs prior to the onset of pronounced capacity degradation, suggesting that the proposed indicator responds sensitively to early changes in interfacial dynamics associated with abnormal charging behavior.

A comparable evolution of γ is observed for the fast-charging dataset reported in Ref. [37], as shown in Fig. 8c and d (see Fig. S9 for more results). Notably, the cycle number range for γ is shorter than that for capacity retention in these figures because the dataset later shifted to milder charging conditions. The charging protocol employed in this dataset consists of multiple CC steps followed by a single CC-CV transition (see Fig. S10), making it structurally analogous to the Type 2 protocol defined in Section 3.4. Accordingly, γ was extracted from the CC stage immediately preceding the CC-CV transition. Despite differences in cell chemistry, charging protocols, and experimental conditions between the fast-charging dataset and the CALCE dataset, the indicator γ exhibits an evolution trend consistent with that obtained from the CALCE dataset (e.g., at the 850th cycle in Fig. 8c and the 734th cycle in Fig. 8d).

To further verify that the observed transition in γ is associated with lithium plating, additional electrochemical analyses were performed on the same datasets. For the CALCE dataset (LiCoO₂/graphite), incremental capacity analysis (ICA, dQ/dV) was conducted. As shown in Fig. S11a-d, characteristic features commonly associated with lithium plating emerge at cycle numbers 460 and 290, which align with the transition points observed in γ . This provides a clear, cycle-resolved confirmation that γ captures the onset of plating in these cells.

For the fast-charging dataset reported by Attia et al. (LiFePO₄/graphite), coulombic efficiency (CE) was analyzed (see Fig. S11e-f). A systematic deviation from near-unity efficiency is observed at the cycles corresponding to the γ transition, indicating that plating has occurred. It should be noted that, due to the fast-charging protocol, lithium plating in this dataset may have already started before the CC-CV segment from which γ is extracted. Therefore, the plating onset identified by the γ transition reflects only the CC charging phase within the CC-CV segment, and may differ from the cycle number corresponding to plating onset inferred from the deviation of CE from 100% over the entire fast-charging protocol. Nevertheless, the clear deviation of CE from 100% at the γ transition cycles confirms that lithium plating occurs, and γ effectively captures the associated electrochemical changes.

The agreement between experimental observations and model predictions is noteworthy. Across both datasets, γ exhibits the same characteristic evolution identified in the mechanistic analysis: a smooth variation during normal aging followed by a pronounced trend transi-

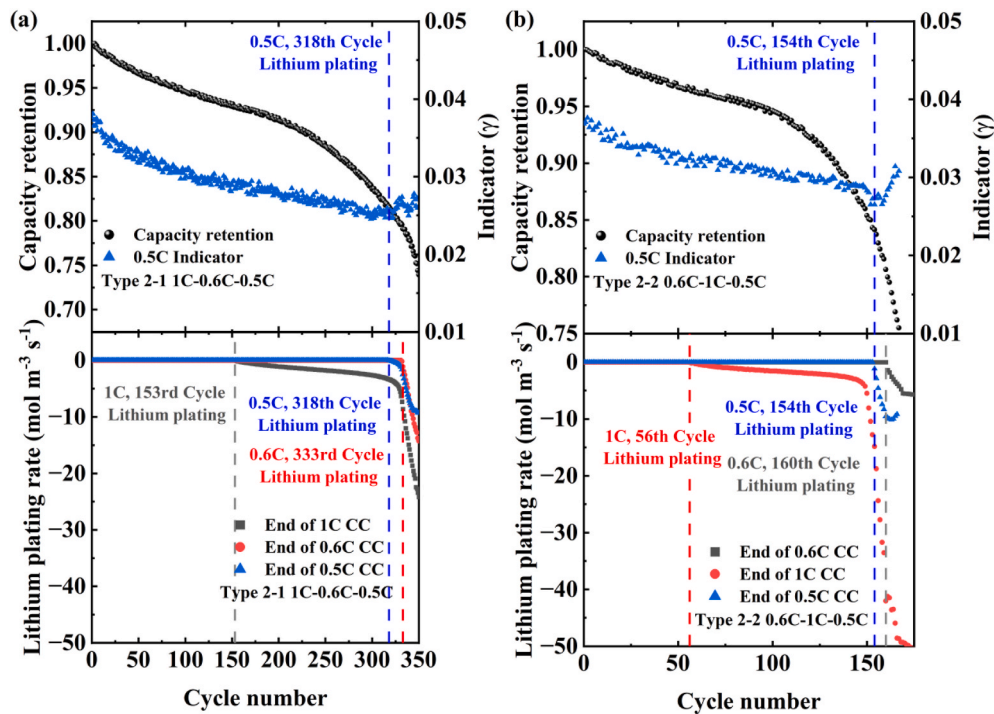


Fig. 7. Identification of lithium plating under multi-step CC charging followed by a single CV step (Type 2). The left and right panels correspond to the Type 2-1 (a) and Type 2-2 (b) charging protocols, respectively. The top panel shows the evolution of capacity retention together with the indicator γ extracted from the 0.5C CC-CV transition, while the bottom panel presents the lithium plating reaction rates at the end of each CC step.

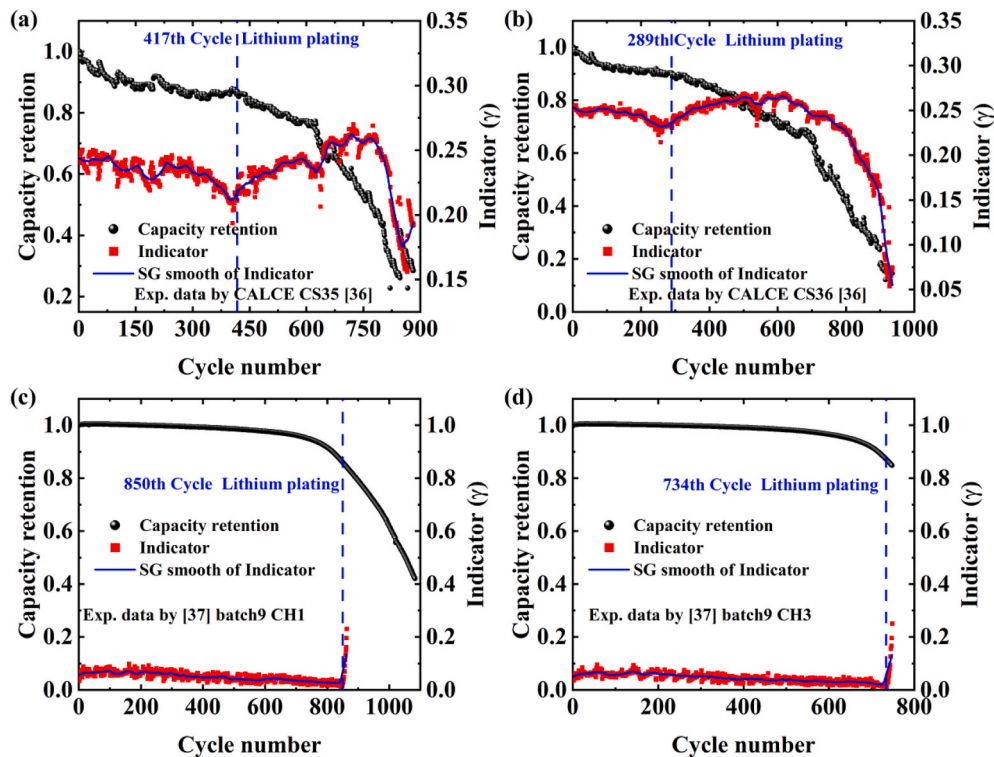


Fig. 8. Evolution of the transient-current indicator γ extracted from open-source experimental datasets. (a,b) Results for selected CALCE cells and (c,d) results for the fast-charging lithium-ion battery dataset reported in Ref. [37]. In all panels, capacity retention is plotted on the primary (left) y-axis, while γ is shown on the secondary (right) y-axis.

tion coinciding with plating-consistent electrochemical signatures. This demonstrates that the proposed transient-current indicator reliably reflects intrinsic electrochemical changes at the CC-CV transition rather

than dataset-specific artifacts.

Overall, the validation results based on independent open-source experimental data demonstrate that the proposed indicator γ is not

only theoretically grounded but also experimentally observable and robust across different cell types and charging conditions. These findings highlight the potential of the proposed method as a low-cost, model-informed diagnostic tool for early lithium plating detection in practical battery management systems.

4. Conclusion

In this work, we developed a simple and robust electrochemical indicator γ based on the transient current response at the CC–CV transition to enable early detection of lithium plating. By combining a mechanism-based DFN model with extensive simulation analysis, we demonstrated that this indicator effectively discriminates between plating-free and plating conditions. Unlike capacity retention analysis which often exhibits a delayed response, or conventional voltage relaxation methods that necessitate prolonged rest periods for thermodynamic equilibrium, the proposed indicator operates directly during the active charging phase. It displays a sensitive inflection point—transitioning from a decreasing to an increasing trend—precisely at the onset of lithium plating. This distinct behavior allows for the identification of plating risks approximately 50 cycles earlier than capacity fade analysis and offers superior real-time sensitivity compared to relaxation-based techniques, providing a critical lead time for active safety intervention.

The diagnostic capability of γ is theoretically grounded in the unique interfacial kinetics occurring at the CC–CV switch. Our mechanistic analysis reveals that the transition involves the rapid suppression of the plating reaction due to reduced polarization, superimposed onto an intercalation current that has been historically suppressed during the CC phase. This abrupt kinetic redistribution creates a signature current decay that is intrinsically linked to the thermodynamic and kinetic state of the anode surface. Furthermore, the indicator consistently identifies plating onset when a plating-free baseline exists, while under immediate-plating high-rate conditions such as 1.5C it should be interpreted as a persistent plating-risk indicator rather than an onset detector.

Finally, the practical applicability of the proposed method was validated using independent open-source experimental datasets, including CALCE and fast-charging data. The agreement between model predictions and experimental results confirms that the indicator captures fundamental electrochemical changes rather than protocol-specific artifacts. Distinct from techniques requiring complex signal injection or interruption of battery operation, our approach relies solely on standard current and voltage data monitored during the charging process. Consequently, it offers a cost-effective, model-informed solution suitable for online integration into commercial battery management systems to enhance the safety and longevity of lithium-ion batteries.

CRedit authorship contribution statement

Lizhi Xiang: Data curation, Formal analysis, Investigation, Writing – original draft. **Binghan Cui:** Writing – review & editing. **Fangmin Wu:** Conceptualization. **Daode Wang:** Conceptualization. **Geping Yin:** Formal analysis, Investigation. **Huo Hua:** Formal analysis, Investigation. **Chunyu Du:** Conceptualization, Funding acquisition, Methodology, Supervision, Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.jpowsour.2026.240403>.

Data availability

Data will be made available on request.

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